24V input on GIGA shield | Local 3.3V display conversion | <=100 mm Type-A 80-way FFC

ribbon

Datei: ribbon.kicad_sch

buffer1

Datei: buffer1.kicad_sch

buffer2

Datei: buffer2.kicad_sch

buffer3

Datei: buffer3.kicad_sch

buffer4

Datei: buffer4.kicad_sch

touch

Datei: touch.kicad_sch

power_input

Datei: power_input.kicad_sch

power12

Datei: power12.kicad_sch

headers

Datei: headers.kicad_sch

Supplies present

power:VBUS

Supplies present

power:VBUS_PANEL

Supplies present

power:VBUS_GIGA

ProtoMate

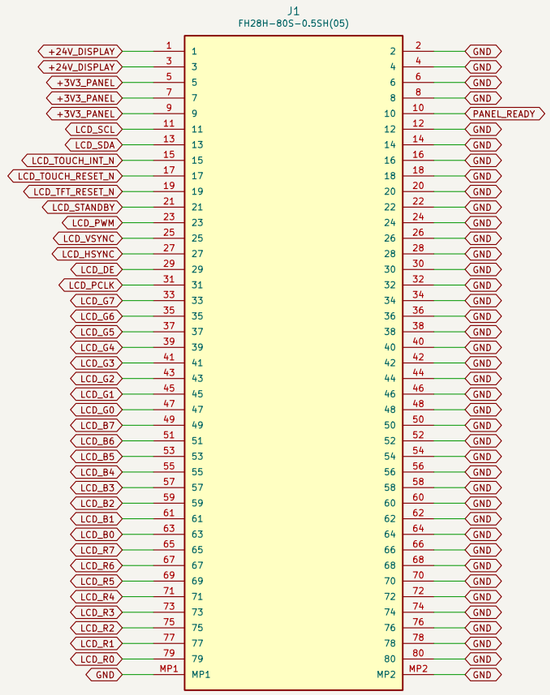
Sheet: /

File: giga-adapter.kicad_sch

Title: giga-adapter / Smartwin display interface

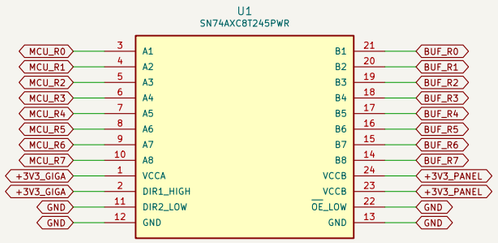
Size: A3	Date: 2026-10-21	**Rev: D**
KiCad E.D.A. 10.0.6		Id: 1/10

80-way Type-A (same-side contacts), 0.5mm pitch, 0.30mm mating thickness. GIGA pin n <-> display socket pin 81-n.



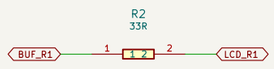
ProtoMate		
Sheet: /ribbon/		
File: ribbon.kicad_sch		
Title: giga-adapter / ribbon		
Size: A3	Date: 2026-10-21	**Rev: D**
KiCad E.D.A. 10.0.6		Id: 2/10

DIR1 = HIGH; DIR2 = LOW; OE = LOW. 33R series termination; 47k input bias. All channels A -> B.

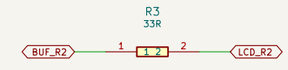




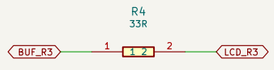
















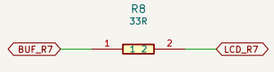




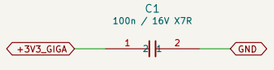


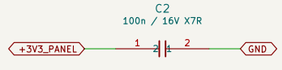












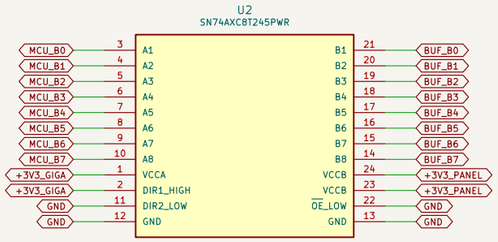
ProtoMate

Sheet: /buffer1/
File: buffer1.kicad_sch

Title: giga-adapter / buffer1

Size: A3	Date: 2026-10-21	**Rev: D**
KiCad E.D.A. 10.0.6		Id: 3/10

Same isolation and termination as U1. Both supplies must be valid for transmission.





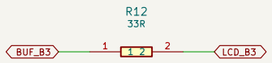




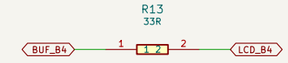




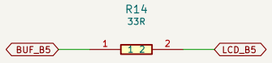












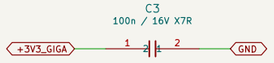


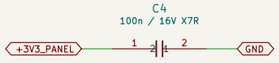












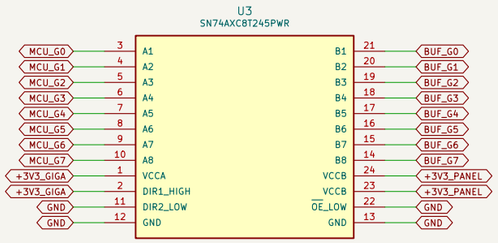
ProtoMate

Sheet: /buffer2/
File: buffer2.kicad_sch

Title: giga-adapter / buffer2

Size: A3	Date: 2026-10-21	**Rev: D**
KiCad E.D.A. 10.0.6		Id: 4/10

RGB24 physical interface. All eight green bits are connected.



























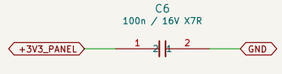












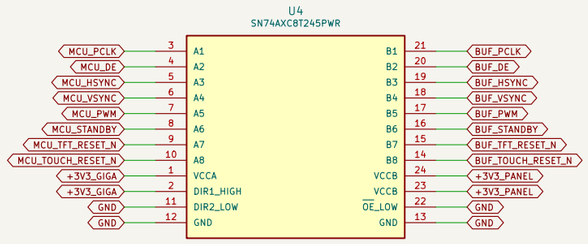
ProtoMate

Sheet: /buffer3/
File: buffer3.kicad_sch

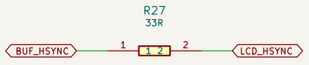
Title: giga-adapter / buffer3

Size: A3	Date: 2026-10-21	**Rev: D**
KiCad E.D.A. 10.0.6		Id: 5/10

PCLK, DE, HSYNC, VSYNC, PWM, STANDBY and active-low resets. PCLK buffered by the same family as RGB.

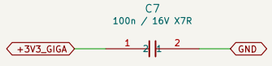
























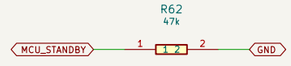














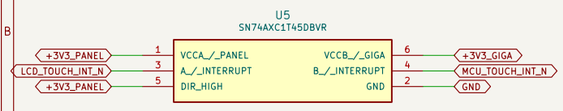
ProtoMate

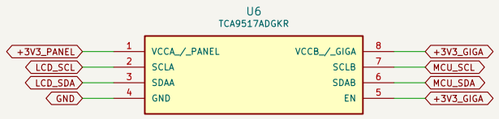
Sheet: /buffer4/
File: buffer4.kicad_sch

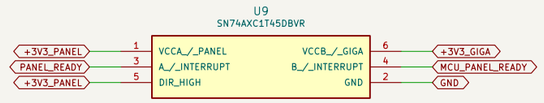
Title: giga-adapter / buffer4

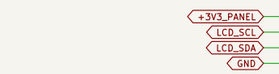
Size: A3	Date: 2026-10-21	**Rev: D**
KiCad E.D.A. 10.0.6		Id: 6/10

TCA9517 A = panel; B = GIGA. GIGA R26/R27 already provide 5.1k pull-ups. AXC1 direction A -> B.











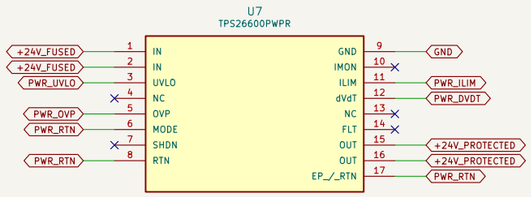
ProtoMate

Sheet: /touch/
File: touch.kicad_sch

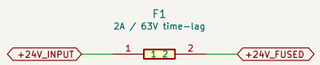
Title: giga-adapter / touch

Size: A3	Date: 2026-10-21	**Rev: D**
KiCad E.D.A. 10.0.6		Id: 7/10

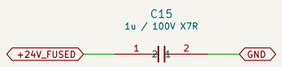
24V nominal input. UVLO ~18.2V; OVP ~28.1V; current limit ~1.49A. RTN is NOT system GND. SHDN floats (internal pull-up).











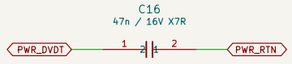


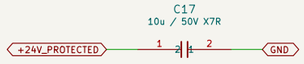














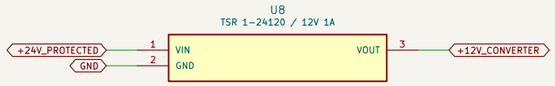
ProtoMate

Sheet: /power_input/
File: power_input.kicad_sch

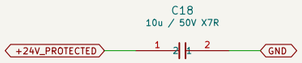
Title: giga-adapter / power_input

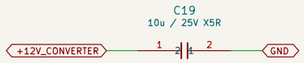
Size: A3	Date: 2026-10-21	**Rev: D**
KiCad E.D.A. 10.0.6		Id: 8/10

U8: 24V -> 12V / 1A. D3 blocks USB-only reverse feed into U8. F2 protects the two parallel 24V ribbon conductors.















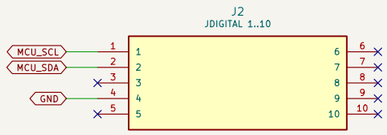
ProtoMate

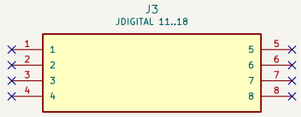
Sheet: /power12/
File: power12.kicad_sch

Title: giga-adapter / power12

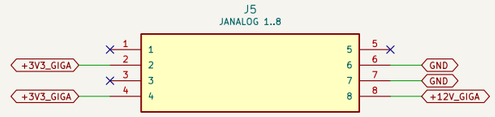
Size: A3	Date: 2026-10-21	**Rev: D**
KiCad E.D.A. 10.0.6		Id: 9/10

NC means no adapter trace. Socket still passes that GIGA contact straight through. Do not populate other header variants.



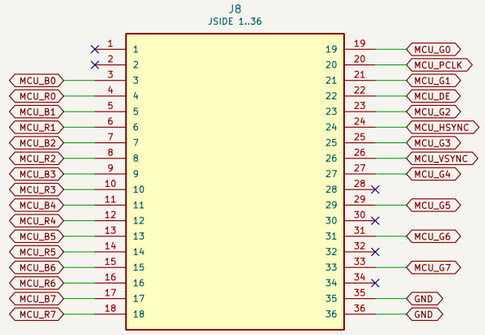


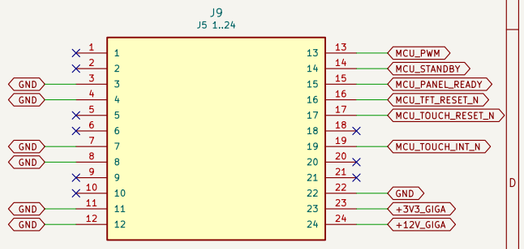












ProtoMate

Sheet: /headers/
File: headers.kicad_sch

Title: giga-adapter / headers

Size: A3	Date: 2026-10-21	**Rev: D**
KiCad E.D.A. 10.0.6		Id: 10/10